CSE 3204
Formal Language, Automata and Computability

Lecture 25
Intractable Problems

The Satisfiability Problem

The *boolean expressions* are built from:

1. Variables whose values are boolean; i.e., they either have the value 1 (true) or 0 (false).
2. Binary operators $\wedge$ and $\vee$, standing for the logical AND and OR of two expressions.
3. Unary operator $\neg$ standing for logical negation.
4. Parentheses to group operators and operands, if necessary to alter the default precedence of operators: $\neg$ highest, then $\wedge$, and finally $\vee$.

The *satisfiability problem* is:

- Given a boolean expression, is it satisfiable?

We shall generally refer to the satisfiability problem as *SAT*. Stated as a language, the problem SAT is the set of (coded) boolean expressions that are satisfiable. Strings that either are not valid codes for a boolean expression or that are codes for an unsatisfiable boolean expression are not in SAT.

A *truth assignment* for a given boolean expression E assigns either true or false to each of the variables mentioned in E. The *value* of expression E given a truth assignment T, denoted $E(T)$, is the result of evaluating E with each variable x replaced by the value $T(x)$ (true or false) that T assigns to x.

A truth assignment T *satisfies* boolean expression E if $E(T) = 1$; i.e., the truth assignment T makes expression E true. A boolean expression E is said to be *satisfiable* if there exists at least one truth assignment T that satisfies E.

NP-Completeness of the SAT Problem

- SAT is in NP
- every language in NP reduces to the SAT problem.
- **Cook's Theorem:** SAT is NP-Complete.

1. Use the nondeterministic ability of an NTM to guess a truth assignment T for the given expression E. If the encoded E is of length n, then $O(n)$ time suffices on a multitape NTM. Note that this NTM has many choices of move, and may have as many as 2^n different ID's reached at the end of the guessing process, where each branch represents the guess of a different truth assignment.

2. Evaluate E for the truth assignment T.

The evaluation can be done easily in $O(n^2)$ time on a multitape NTM. Thus, the entire recognition of SAT by the multitape NTM takes $O(n^2)$ time. Converting to a single-tape NTM may square the amount of time, so $O(n^4)$ time suffices on a single-tape NTM.

Need to prove that for any L in NP, there is a polynomial time reduction of L to SAT.

NP-Completeness of the SAT Problem

is some single-tape NTM M and a polynomial $p(n)$ such that M takes no more than $p(n)$ steps on an input of length n, along any branch. Further, the restrictions of Theorem 8.12, which we proved for DTM's, can be proved in the same way for NTM's. Thus, we may assume that M never writes a blank, and never moves its head left of its initial head position.

Thus, if M accepts an input w, and $|w| = n$, then there is a sequence of moves of M such that:

1. α_0 is the initial ID of M with input w.
2. $\alpha_0 \vdash \alpha_1 \vdash \cdots \vdash \alpha_k$, where $k \leq p(n)$.
3. α_k is an ID with an accepting state.
4. Each α_i consists of nonblanks only (except if α_i ends in a state and a blank), and extends from the initial head position — the leftmost input symbol — to the right.

NP-Completeness of the SAT Problem

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NP-Completeness of the SAT Problem

a) Each α_i can be written as a sequence of symbols $X_{i0}X_{i1}\cdots X_{i,p(n)}$. One of these symbols is a state, and the others are tape symbols. As always,

b) To describe the sequence of ID's in terms of boolean variables, we create variable y_{ijA} to represent the proposition that $X_{ij} = A$. Here, i and j are each integers in the range 0 to $p(n)$, and A is either a tape symbol or a state.

c) We express the condition that the sequence of ID's represents acceptance of an input w by writing a boolean expression that is satisfiable if and only if M accepts w by a sequence of at most $p(n)$ moves. The satisfying assignment will be the one that "tells the truth" about the ID's; that is, y_{ijA} will be true if and only if $X_{ij} = A$. To make sure that the polynomial-time reduction of $L(M)$ to SAT is correct, we write this expression so that it says the computation:

i. *Starts right.* That is, the initial ID is q_0w followed by blanks.

ii. *Next move is right* (i.e., the move correctly follows the rules of the TM). That is, each subsequent ID follows from the previous by one of the possible legal moves of M.

iii. *Finishes right.* That is, there is some ID that is an accepting state.

- First, we have specified ID's to end when the infinite tail of blanks begin. However, it is more convenient when simulating a polynomial-time computation to think of all ID's as having the same length, $p(n) + 1$. Thus, a tail of blanks may be present in an ID.

- Second, it is convenient to assume that all computations continue for exactly $p(n)$ moves [and therefore have $p(n) + 1$ ID's], even if acceptance occurs earlier. We therefore allow each ID with an accepting state to be its own successor. That is, if α has an accepting state, we allow a "move" $\alpha \vdash \alpha$. Thus, we can assume that if there is an accepting computation, then $\alpha_{p(n)}$ will have an accepting ID, and that is all we have to check for the condition "finishes right."

NP-Completeness of the SAT Problem

ID	0	1	...				...	$p(n)$
α_0	X_{00}	X_{01}						$X_{0,p(n)}$
α_1	X_{10}	X_{11}						$X_{1,p(n)}$
α_i				$X_{i,j-1}$	$X_{i,j}$	$X_{i,j+1}$		
α_{i+1}				$X_{i+1,j-1}$	$X_{i+1,j}$	$X_{i+1,j+1}$		
$\alpha_{p(n)}$	$X_{p(n),0}$	$X_{p(n),1}$						$X_{p(n),p(n)}$

Figure 10.4: Constructing the array of cell/ID facts

Let us now give an algorithm to construct from M and w a boolean expression $E_{M,w}$. The overall form of $E_{M,w}$ is $U \wedge S \wedge N \wedge F$, where S, N, and F are expressions that say M starts, moves, and finishes right, and U says there is a unique symbol in each cell.

Unique

U is the logical AND of all terms of the form $\neg(y_{ij\alpha} \wedge y_{ij\beta})$, where $\alpha \neq \beta$. Note the number of these terms is $O\big(p^2(n)\big)$.

Starts Right

X_{00} must be the start state q_0 of M, X_{01} through X_{0n} must be w (where n is the length of w), and the remaining X_{0j} must be the blank, B. That is, if $w = a_1 a_2 \cdots a_n$, then:

$$S = y_{00q_0} \wedge y_{01a_1} \wedge y_{02a_2} \wedge \cdots \wedge y_{0na_n} \wedge y_{0,n+1,B} \wedge y_{0,n+2,B} \wedge \cdots \wedge y_{0,p(n),B}$$

Surely, given the encoding of M and given w, we can write S in $O\big(p(n)\big)$ time on a second tape of a multitape TM.

NP-Completeness of the SAT Problem

Finishes Right

Since we assume that an accepting ID repeats forever, acceptance by M is the same as finding an accepting state in $\alpha_{p(n)}$. Remember that we assume M is an NTM that, if it accepts, does so within $p(n)$ steps. Thus, F is the OR of expressions F_j, for $j = 0, 1, \ldots, p(n)$, where F_j says that $X_{p(n),j}$ is an accepting state. That is, F_j is $y_{p(n),j,a_1} \vee y_{p(n),j,a_2} \vee \cdots \vee y_{p(n),j,a_k}$, where $a_1, a_2, \ldots, a_k$ are all the accepting states of M. Then, $F = F_0 \vee F_1 \vee \cdots \vee F_{p(n)}$.

Each F_i uses a constant number of symbols, depending on M but not on the length n of its input w. Thus, F has length $O(n)$. More importantly, the time to write F, given an encoding of M and the input w is polynomial in n; actually, F can be written in $O\big(p(n)\big)$ time on a multitape TM.

Next Move is Right

Assuring that the moves of M are correct is by far the most complicated part. The expression N will be the AND of expressions N_i, for $i = 0, 1, \ldots, p(n) - 1$, and each N_i will be designed to assure that ID α_{i+1} is one of the ID's that M allows to follow α_i. To begin the explanation of how to write N_i, observe symbol $X_{i+1,j}$ in Fig. 10.4. We can always determine $X_{i+1,j}$ from:

1. The three symbols above it: $X_{i,j-1}$, X_{ij}, and $X_{i,j+1}$, and
2. If one of these symbols is the state of α_i, then the particular choice of move by the NTM M.

We shall write N_i as the $\wedge$ of expressions $A_{ij} \vee B_{ij}$, where $j = 0, 1, \ldots, p(n)$.

NP-Completeness of the SAT Problem

- Expression A_{ij} says that:

  a) The state of α_i is at position j (i.e., X_{ij} is the state), and

  b) There is a choice of move of M, where X_{ij} is the state and $X_{i,j+1}$ is the symbol scanned, such that this move transforms the sequence of symbols $X_{i,j-1}X_{ij}X_{i,j+1}$ into $X_{i+1,j-1}X_{i+1,j}X_{i+1,j+1}$. Note that if X_{ij} is an accepting state, there is the "choice" of making no move at all, so all subsequent ID's are the same as the one that first led to acceptance.

- Expression B_{ij} says that:

  a) The state of α_i is not at position j (i.e., X_{ij} is not a state, and

  b) If the state of α_i is not adjacent to position j (i.e., $X_{i,j-1}$ and $X_{i,j+1}$ are not states either), then $X_{i+1,j} = X_{ij}$.

B_{ij} is the easier to write. Let $q_1, q_2, \ldots, q_m$ be the states of M, and let $Z_1, Z_2, \ldots, Z_r$ be the tape symbols. Then:

$$
\begin{aligned}
B_{ij} \quad = \quad & (y_{i,j-1,q_1} \vee y_{i,j-1,q_2} \vee \cdots \vee y_{i,j-1,q_m}) \vee \\
& (y_{i,j+1,q_1} \vee y_{i,j+1,q_2} \vee \cdots \vee y_{i,j+1,q_m}) \vee \\
& \Big((y_{i,j,Z_1} \vee y_{i,j,Z_2} \vee \cdots \vee y_{i,j,Z_r}) \wedge \\
& \big((y_{i,j,Z_1} \wedge y_{i+1,j,Z_1}) \vee (y_{i,j,Z_2} \wedge y_{i+1,j,Z_2}) \vee \cdots \vee (y_{i,j,Z_r} \wedge y_{i+1,j,Z_r}) \big) \Big)
\end{aligned}
$$

NP-Completeness of the SAT Problem

Now, let us consider the expressions A_{ij}. These expressions reflect all possible relationships among the 2×3 rectangle of symbols in the array of Fig. 10.4: $X_{i,j-1}$, X_{ij}, $X_{i,j+1}$, $X_{i+1,j-1}$, $X_{i+1,j}$, and $X_{i+1,j+1}$. An assignment of symbols to each of these six variables is *valid* if:

1. X_{ij} is a state, but $X_{i,j-1}$ and $X_{i,j+1}$ are tape symbols.

2. There is a move of M that explains how $X_{i,j-1}X_{ij}X_{i,j+1}$ becomes

$$X_{i+1,j-1}X_{i+1,j}X_{i+1,j+1}$$

There are thus a finite number of assignments of symbols to the six variables that are valid. Let A_{ij} be the OR of terms, one term for each set of six variables that form a valid assignment.

For instance, suppose that one move of M comes from the fact that $\delta(q, A)$ contains (p, C, L). Let D be some tape symbol of M. Then one valid assignment is $X_{i,j-1}X_{ij}X_{i,j+1} = DqA$ and $X_{i+1,j-1}X_{i+1,j}X_{i+1,j+1} = pDC$. Notice how this assignment reflects the change in ID that is caused by making this move of M. The term that reflects this possibility is

$$y_{i,j-1,D} \wedge y_{i,j,q} \wedge y_{i,j+1,A} \wedge y_{i+1,j-1,p} \wedge y_{i+1,j,D} \wedge y_{i+1,j+1,C}$$

If, instead, $\delta(q, A)$ contains (p, C, R) (i.e., the move is the same, but the head moves right), then the corresponding valid assignment is $X_{i,j-1}X_{ij}X_{i,j+1} = DqA$ and $X_{i+1,j-1}X_{i+1,j}X_{i+1,j+1} = DCp$. The term for this assignment is

$$y_{i,j-1,D} \wedge y_{i,j,q} \wedge y_{i,j+1,A} \wedge y_{i+1,j-1,D} \wedge y_{i+1,j,C} \wedge y_{i+1,j+1,p}$$

A_{ij} is the OR of all valid terms. In the special cases $j = 0$ and $j = p(n)$, we must make certain modifications to reflect the nonexistence of the variables y_{ijZ} for $j < 0$ or $j > p(n)$, as we did for B_{ij}. Finally,

$$N_i = (A_{i0} \vee B_{i0}) \wedge (A_{i1} \vee B_{i1}) \wedge \cdots \wedge (A_{i,p(n)} \vee B_{i,p(n)})$$

and then

NP-Completeness of the SAT Problem

$$N = N_0 \wedge N_1 \wedge \cdots \wedge N_{p(n)-1}$$

Although A_{ij} and B_{ij} can be very large if M has many states and/or tape symbols, their size is actually a constant as far as the length of input w is concerned; that is, their size is independent of n, the length of w. Thus, the length of N_i is $O\big(p(n)\big)$, and the length of N is $O\big(p^2(n)\big)$. More importantly, we can write N on a tape of a multitape TM in an amount of time that is proportional to its length, and that amount of time is polynomial in n, the length of w.

Conclusion of the Proof of Cook's Theorem

Although we have described the construction of the expression

$$E_{M,w} = U \wedge S \wedge N \wedge F$$

as a function of both M and w, observe that it is only the "starts right" part S that depends on w, and it does so in a simple way (w is on the tape of the initial ID). The other parts, N and F, depend on M and on n, the length of w, only.

Thus, for any NTM M that runs in some polynomial time $p(n)$, we can devise an algorithm that takes an input w of length n, and produces $E_{M,w}$. The running time of this algorithm on a multitape, deterministic TM is $O\big(p^2(n)\big)$, and that multitape TM can be converted to a single-tape TM that runs in time $O\big(p^4(n)\big)$. The output of this algorithm is a boolean expression $E_{M,w}$ that is satisfiable if and only if M accepts w within $p(n)$ moves. $\square$

- Section 10.2 of Chapter 10.